

Combs, Causality and Contractions in Atomic Markov Categories

Dario Stein and **Márk Széles**

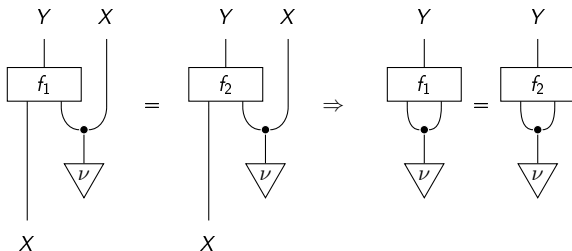
Radboud University, Nijmegen

ACT 2024

Related work

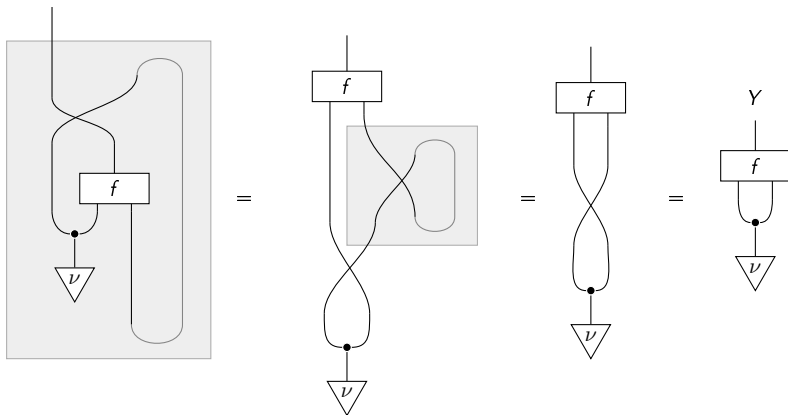
- Mathematical Framework for Causally Structured Dilations and its Relation to Quantum Self-Testing (*Houghton-Larsen, 2021*)
- Coend optics for quantum combs (*Hefford & Comfort, 2022*)
- Absolute continuity, supports and idempotent splitting in categorical probability (*Fritz, Gonda, Lorenzin, Perrone, Stein, 2023*)

Contraction identities



Contraction identities

Contraction identities hold in traced monoidal categories:



Contraction identities in BorelStoch

Counterexample

Contraction identities fail in BorelStoch

Corollary

BorelStoch cannot be embedded into any traced monoidal category

Contraction identities in BorelStoch

Counterexample

Contraction identities fail in BorelStoch

Corollary

BorelStoch cannot be embedded into any traced monoidal category

Why do we care? Combs and causality!

- $\text{FinStoch} \rightarrow \text{Mat}(\mathbb{R}^+)$ (*Jacobs, Kissinger & Zanasi, 2021*)
- $\text{BorelStoch} \rightarrow ???$

Contraction identities in BorelStoch

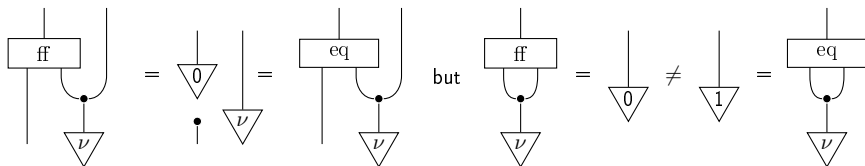
Let $X = [0, 1]$ and $\nu : I \rightarrow X$ be the Lebesgue measure, and $\text{ff}, \text{eq} : X \otimes X \rightarrow \{0, 1\}$ the measurable functions

$$\text{ff}(x, y) = 0, \quad \text{eq}(x, y) = \begin{cases} 1, & x = y \\ 0, & \text{otherwise} \end{cases}$$

Contraction identities in BorelStoch

Let $X = [0, 1]$ and $\nu : I \rightarrow X$ be the Lebesgue measure, and $\text{ff}, \text{eq} : X \otimes X \rightarrow \{0, 1\}$ the measurable functions

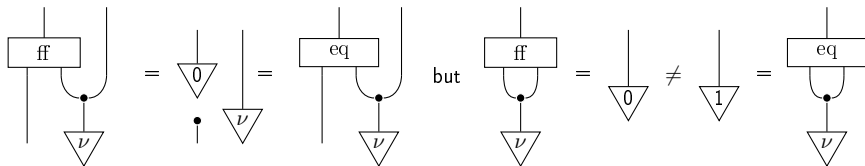
$$\text{ff}(x, y) = 0, \quad \text{eq}(x, y) = \begin{cases} 1, & x = y \\ 0, & \text{otherwise} \end{cases}$$



Contraction identities in BorelStoch

Let $X = [0, 1]$ and $\nu : I \rightarrow X$ be the Lebesgue measure, and $\text{ff}, \text{eq} : X \otimes X \rightarrow \{0, 1\}$ the measurable functions

$$\text{ff}(x, y) = 0, \quad \text{eq}(x, y) = \begin{cases} 1, & x = y \\ 0, & \text{otherwise} \end{cases}$$

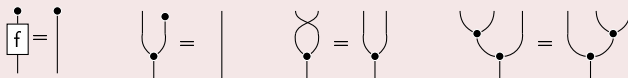


Source of the problem: BorelStoch is not atomic!

Markov categories

Definition (*Fritz, 2020*)

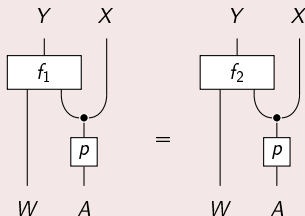
A **Markov category** is a symmetric monoidal $(\mathcal{C}, \otimes, I)$ with $\Delta_X : X \rightarrow X \otimes X$, $\text{del}_X : X \rightarrow I$ such that



Almost sure equality and absolute continuity

Definition (Fritz et al, 2023)

- f_1 is p -almost surely equal to f_2 if



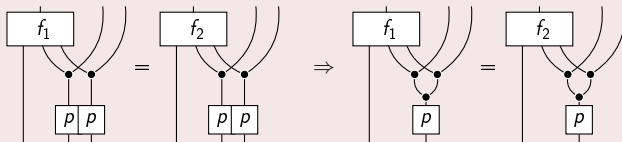
- p is absolutely continuous w.r.t q ($p \ll q$) if

$$(f_1 =_q f_2 \Rightarrow f_1 =_p f_2)$$

Atomicity

Definition (*Fritz et al, 2023*)

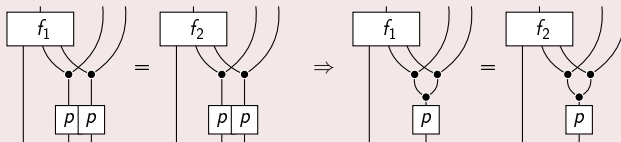
■ $p : A \rightarrow X$ is **atomic** if $\Delta_X \circ p \ll p \otimes p$



Atomicity

Definition (*Fritz et al, 2023*)

- $p : A \rightarrow X$ is **atomic** if $\Delta_X \circ p \ll p \otimes p$

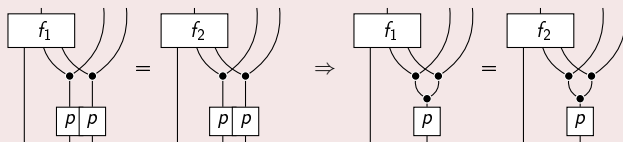


- A Markov category is **atomic** if every morphism is atomic

Atomicity

Definition (Fritz et al, 2023)

- $p : A \rightarrow X$ is **atomic** if $\Delta_X \circ p \ll p \otimes p$



- A Markov category is **atomic** if every morphism is atomic

Example: FinStoch is atomic ($\text{supp}(\Delta \circ p) \subseteq \text{supp}(p \otimes p)$)

Atomicity fails in BorelStoch

In BorelStoch $p : A \rightarrow X$ is atomic $\iff$ p is **completely atomic** in the sense of measure theory:

$$p(\mathcal{A}|a) = 1$$

where

$$\mathcal{A} = \{x \in X \mid \exists a \in A. p(\{x\}|a) > 0\}$$

The Lebesgue-measure is not completely atomic: counterexample from before

Atomicity fails in BorelStoch

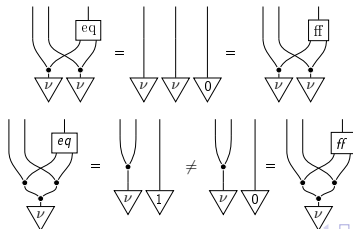
In BorelStoch $p : A \rightarrow X$ is atomic $\iff$ p is **completely atomic** in the sense of measure theory:

$$p(\mathcal{A}|a) = 1$$

where

$$\mathcal{A} = \{x \in X \mid \exists a \in A. p(\{x\}|a) > 0\}$$

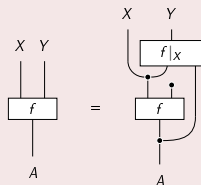
The Lebesgue-measure is not completely atomic: counterexample from before



Conditionals and cancellativity

Definition

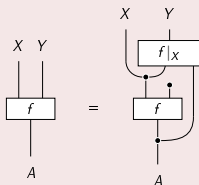
- A Markov category has **conditionals** if



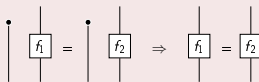
Conditionals and cancellativity

Definition

- A Markov category has **conditionals** if



- A Markov category is **cancellative** if



Non-signalling morphisms

Definition

f is **non-signalling** from W to W if

The diagram shows an equality between two morphisms. On the left, a box labeled f has two input wires at the bottom labeled X and W , and two output wires at the top. The right output wire is labeled W and has a black dot on it. On the right, the same box is labeled f_s and has only one input wire at the bottom labeled X . To the right of the f_s box is a separate wire labeled W with a black dot on it. An equals sign is placed between the two boxes.

Non-signalling morphisms

Definition

f is **non-signalling** from W to W if

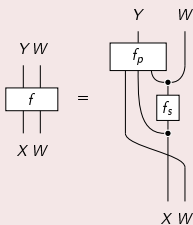
$$\begin{array}{c} W \\ \bullet \\ \boxed{f} \\ \begin{array}{cc} \downarrow & \downarrow \\ X & W \end{array} \end{array} = \begin{array}{c} W \\ \boxed{f_s} \\ \downarrow \\ X \end{array} \quad \begin{array}{c} \bullet \\ \downarrow \\ W \end{array}$$

We hope to define a trace operation for non-signalling morphisms

Non-signalling morphisms

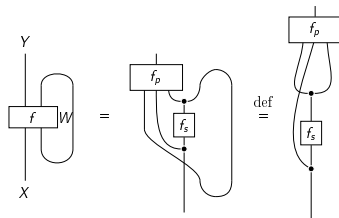
Proposition

If $\mathcal{C}$ has conditionals $f : X \otimes W \rightarrow Y \otimes W$ is non-signalling iff



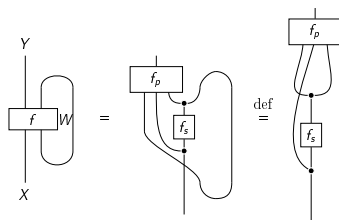
Causal traces in atomic Markov categories

Define the **causal trace** as



Causal traces in atomic Markov categories

Define the **causal trace** as

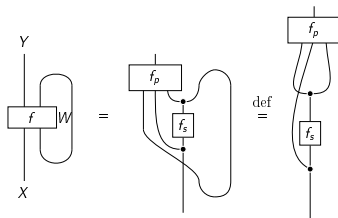


Question

Is this definition independent of the choice of disintegration?

Causal traces in atomic Markov categories

Define the **causal trace** as



Question

Is this definition independent of the choice of disintegration?

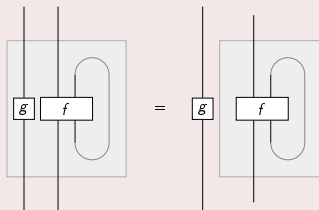
Theorem: yes, if C is atomic

Properties of the causal trace

Proposition (strengthening)

If $f : X \otimes W \rightarrow Y \otimes W$ is traceable then

$g \otimes f : X' \otimes X \otimes W \rightarrow Y' \otimes Y \otimes W$ is traceable, and

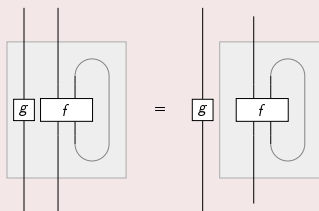


Properties of the causal trace

Proposition (strengthening)

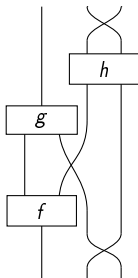
If $f : X \otimes W \rightarrow Y \otimes W$ is traceable then

$g \otimes f : X' \otimes X \otimes W \rightarrow Y' \otimes Y \otimes W$ is traceable, and

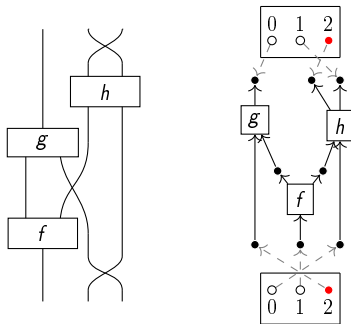


Properties of the causal trace are weaker than the partially traced category axioms (*Haghverdi & Scott, 2005*)

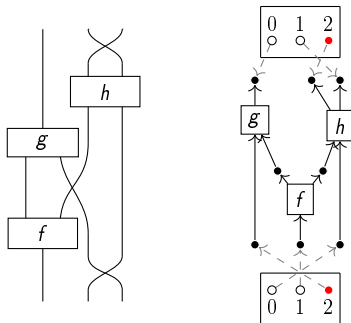
Causal traces in free Markov categories



Causal traces in free Markov categories



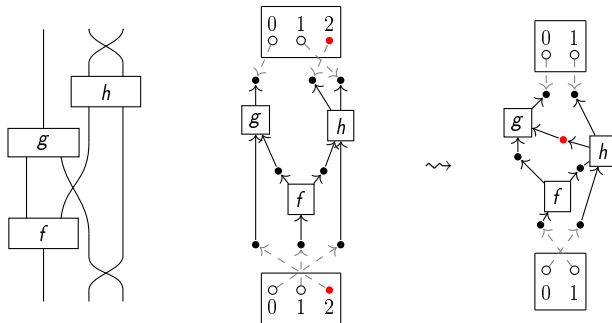
Causal traces in free Markov categories



Proposition

$t : n \otimes w \rightarrow m \otimes w$ is non-signalling iff there are no directed paths from input ports in w to output ports in w

Causal traces in free Markov categories



Proposition

$t : n \otimes w \rightarrow m \otimes w$ is non-signalling iff there are no directed paths from input ports in w to output ports in w

Contraction identities, formally

Definition

$\mathcal{C}$ satisfies all contraction identities if for all $C_1, C_2 : m \otimes w \rightarrow n \otimes w$ non-signalling in a free Markov category over any signature Σ , and all interpreting functors $\llbracket - \rrbracket : \text{FreeMarkov}_\Sigma \rightarrow \mathcal{C}$, we have that if $\llbracket C_1 \rrbracket = \llbracket C_2 \rrbracket$, then $\llbracket \text{contr}^w(C_1) \rrbracket = \llbracket \text{contr}^w(C_2) \rrbracket$.

Contraction identities, formally

Definition

$\mathcal{C}$ satisfies all contraction identities if for all $C_1, C_2 : m \otimes w \rightarrow n \otimes w$ non-signalling in a free Markov category over any signature Σ , and all interpreting functors $\llbracket - \rrbracket : \text{FreeMarkov}_\Sigma \rightarrow \mathcal{C}$, we have that if $\llbracket C_1 \rrbracket = \llbracket C_2 \rrbracket$, then $\llbracket \text{contr}^w(C_1) \rrbracket = \llbracket \text{contr}^w(C_2) \rrbracket$.

Proposition

Cancellative atomic Markov categories with conditionals satisfy all contraction identities

Contraction identities, formally

Definition

$\mathcal{C}$ satisfies all contraction identities if for all $C_1, C_2 : m \otimes w \rightarrow n \otimes w$ non-signalling in a free Markov category over any signature Σ , and all interpreting functors $\llbracket - \rrbracket : \text{FreeMarkov}_\Sigma \rightarrow \mathcal{C}$, we have that if $\llbracket C_1 \rrbracket = \llbracket C_2 \rrbracket$, then $\llbracket \text{contr}^w(C_1) \rrbracket = \llbracket \text{contr}^w(C_2) \rrbracket$.

Proposition

Cancellative atomic Markov categories with conditionals satisfy all contraction identities

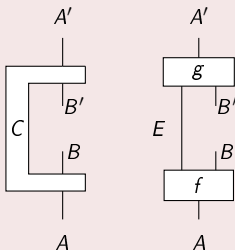
Conjecture

Free Markov categories are atomic, but do not have conditionals

Combs

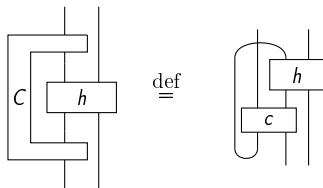
Definition

Extensionally, a **comb** is a non-signalling morphism (as on the left).
Intentionally, it is a pair of morphisms (as on the right).



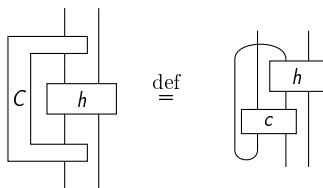
Comb insertion and causal traces

Comb insertion is an instance of causal trace:



Comb insertion and causal traces

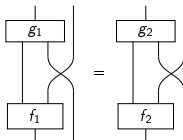
Comb insertion is an instance of causal trace:



Jacobs, Kissinger & Zanasi (2021) used combs for a causal inference problem. Their use of an embedding $\text{FinStoch} \rightarrow \text{Mat}(\mathbb{R}^+)$ is not necessary, thanks to causal traces!

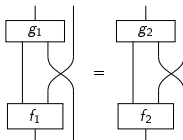
Notions of comb equivalence

- Extensional equivalence:



Notions of comb equivalence

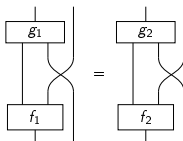
- Extensional equivalence:



- Contextual equivalence: All comb insertions are equal

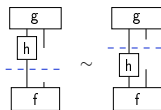
Notions of comb equivalence

- Extensional equivalence:



- Contextual equivalence: All comb insertions are equal
- Optic equivalence: if they are identified as elements of the coend

$$\int^E C(A, E \otimes B) \times C(E \otimes B', A')$$



Results on comb equivalence

Proposition (Hefford & Comfort 2022)

- Optic equivalence $\Rightarrow$ Contextual equivalence $\Rightarrow$ Extensional equivalence
- All three notions of comb equivalence in compact closed and Cartesian categories

Results on comb equivalence

Proposition (Hefford & Comfort 2022)

- Optic equivalence $\Rightarrow$ Contextual equivalence $\Rightarrow$ Extensional equivalence
- All three notions of comb equivalence in compact closed and Cartesian categories

Proposition

- Causal traces imply Extensional $\iff$ Contextual
- Further structure (universal dilations) implies that all notions of comb equivalence coincide (e.g. in FinStoch)

Conclusion

- We defined a notion of causal trace, and show that their existence in a Markov category implies all contraction identities
- We recognised the role of the atomicity axiom in the theory of causal traces
- We gave new sufficient conditions for equivalence of comb equivalences in Markov-categories
- We generalised known results about comb equivalence